RAG Learning

B1.3 Respiration

OCR A GCSE Biology (J247) Gateway — Topic B1: Cell-level systems

1. What is Respiration?

Respiration is the chemical process that takes place in **every cell of every living organism**, transferring energy from glucose to a usable form: **ATP** (adenosine triphosphate). It is an **exothermic** reaction — energy is released, often as heat. Do not confuse respiration with breathing! Breathing is the movement of air in and out of the lungs. Respiration is the cellular process that uses oxygen and glucose.

Energy released by respiration is used by the cell for:

- Building larger molecules from smaller ones (e.g. proteins from amino acids)
- Muscle contraction
- Active transport across cell membranes
- Maintaining body temperature in mammals and birds (warm-blooded animals)
- Keeping the heart and other organs working

2. Aerobic Respiration

Aerobic respiration uses **oxygen** to break down glucose, releasing the maximum amount of energy. It occurs in the **mitochondria**.

$$\text{Word: glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} \ (+\ \text{energy})$$

Symbol: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (+ energy / ATP)

Aerobic Respiration in Mitochondria

Releases the maximum energy from each glucose molecule.

3. Anaerobic Respiration

Anaerobic respiration happens when there is **not enough oxygen** to keep up with energy demand (e.g. during intense exercise). It releases far less energy per molecule of glucose than aerobic respiration because glucose is not fully broken down.

In animals (incl. humans)

glucose → lactic acid (+ a small amount of energy)

Lactic acid builds up in muscles, causing fatigue and a burning sensation. After intense exercise, the body needs to take in **extra oxygen** to break down the lactic acid (the **oxygen debt**). This is why you continue to breathe heavily after stopping exercise.

In plants and yeast — fermentation

glucose → ethanol + carbon dioxide (+ a small amount of energy)

This anaerobic process in yeast is called **fermentation**. It is used commercially to make:

- **Bread** — CO_2 released by yeast makes the dough rise. The ethanol evaporates during baking.
- **Beer / wine** — ethanol gives the alcoholic drink its alcohol content. The CO_2 produces fizz.

	Aerobic respiration	Anaerobic respiration (animal)	Anaerobic respiration (yeast)
Oxygen?	Yes	No	No
Products	CO_2 + H_2O	Lactic acid	Ethanol + CO_2
Energy released per glucose	~30 ATP (large)	~2 ATP (small)	~2 ATP (small)
Where	Mitochondria	Cytoplasm	Cytoplasm
Glucose fully broken down?	Yes	No	No

4. Exercise — Heart Rate, Breathing Rate, Oxygen Debt

During exercise:

- **Heart rate increases** → more blood pumped per minute → more oxygen and glucose delivered to muscles, more CO_2 removed.
- **Breathing rate and depth increase** → more oxygen taken in, more CO_2 removed.
- **Muscles use more oxygen** for aerobic respiration. If demand exceeds supply, anaerobic respiration begins.

> **Oxygen debt**: the extra oxygen needed AFTER exercise to break down the lactic acid that built up during anaerobic respiration. Lactic acid is broken down (in the liver) to CO_2 and water.

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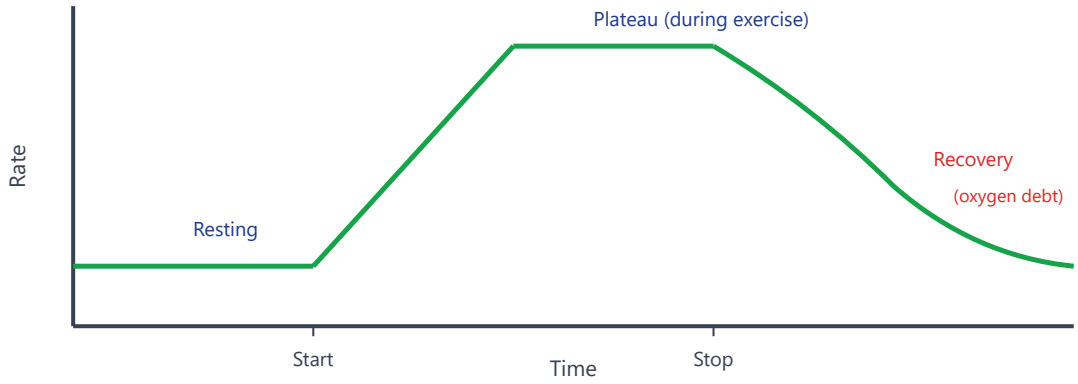
Heart/Breathing Rate Before, During and After Exercise



5. Metabolism and Metabolic Rate

> **Metabolism** = the sum of all chemical reactions occurring in cells (including respiration).
> **Metabolic rate** = the rate at which these reactions take place — usually measured by oxygen consumption or heat production.

Metabolic rate is affected by:

- **Activity** — exercise increases metabolic rate.
- **Body temperature** — fever raises metabolic rate.
- **Genetics, age, sex, body composition** — more muscle mass = higher BMR.
- **Hormones** e.g. thyroxine increases metabolic rate.

6. Required Practical Link

Linked PAG: **investigating the rate of respiration in living organisms**. Often uses germinating peas / small invertebrates with a **respirometer** measuring O_2 consumption (a coloured liquid moves towards the organisms as O_2 is used). See PAG notes for full method.

> **Exam Tip:** Common exam mistake: writing "energy is produced by respiration". Respiration **transfers** or **releases** energy from glucose. Energy cannot be created or destroyed.

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